ZOO513PAnimal Diversity: Chordates (Concised) by M.Kashif (03064963933)

This practical manual explores the diverse world of chordates, covering everything from primitive hemichordates to complex mammals. Each section examines the classification, habitat, feeding habits, and reproductive strategies of different chordate groups, providing students with comprehensive knowledge of animal diversity and evolutionary adaptations.

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Study of Hemichordates

Hemichordates are marine animals commonly called Acorn Worms. Initially grouped under Phylum Chordata, research has shown they lack the notochord and postanal tail characteristic of chordates, resulting in their classification as a separate phylum.

Habitat & Characteristics

- Burrowers in sand or mud, or live in algae or beneath stones
- Body divided into anterior proboscis, short collar, and long trunk
- Size varies from millimeters to 1.5 meters (Balanoglossus)
- Feed on small organic particles as filter feeders or substrate eaters

Classification

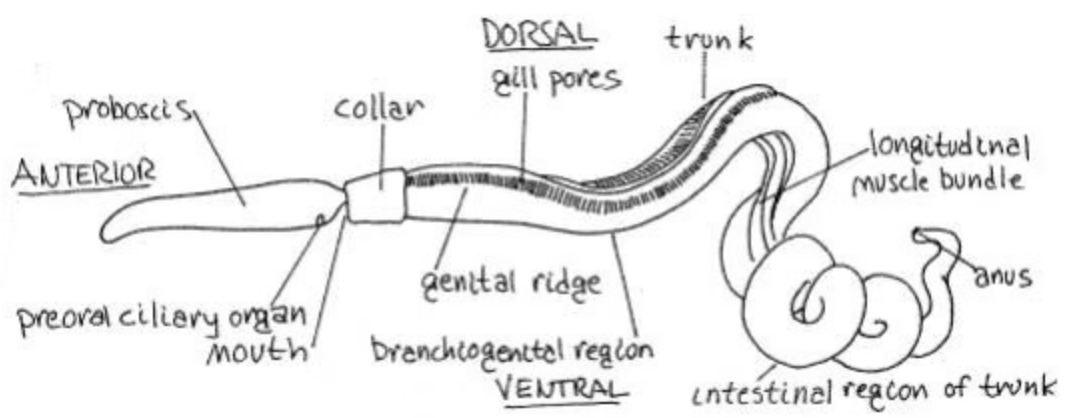
- **Enteropneusta:** Includes acorn worms with vermiform bodies found on sandy beaches in warm climates
- **Pterobranchia:** Tube-dwelling deep sea bottom dwellers that attach to other organisms

Anatomical Features

Acorn worms lack a central nervous system, with nerve tissue concentrated in the collar. The circulatory system includes a contractile heart-like vesicle, blood vessels, and sinuses. The pharynx may be perforated by numerous paired gill slits.

Reproduction

Sexes may be separate or united. Reproduction is mostly sexual with external fertilization. Development is typically indirect, with larvae called tornariae that swim using cilia, though some species exhibit direct development.



Study of Urochordates

Urochordates, also known as Tunicates or "sea squirts," are marine filter-feeding invertebrates that belong to the subphylum Cephalochordates. They feed on planktonic particles and are found in seawater environments worldwide.

Body Structure

The adult tunicate body is sack-shaped with two siphons through which water enters and exits. The body is covered with a thick tunic containing cells, blood vessels, and cellulose - a carbohydrate unusual in animals but abundant in plants.

Larval Stage

Many tunicates have a free-swimming "tadpole larva" that exhibits all chordate characteristics: notochord, dorsal nerve cord, pharyngeal slits, and post-anal tail. After attaching to a substrate, it loses its tail and its nervous system largely disintegrates.

Distribution

Tunicates are distributed throughout the oceans from polar regions to the tropics. Free-swimming forms are found as plankton, while sessile forms grow on solid surfaces like wharf piles, ship hulls, rocks, and shells of various sea creatures.

Nervous System

When the larva metamorphoses into an adult, the original nervous system and sensory organs degenerate, leaving a single ganglion between the oral and atrial openings. Nerves grow to various body organs from this ganglion.

Study of Cephalochordates

Cephalochordates, also called Acrania, are small, fishlike marine invertebrates that are probably the closest living relatives of vertebrates. The name "cephalochordate" comes from Greek (kephale, "head") because the notochord extends from near the tail tip well into the anterior body.

External Features

Lancelets are streamlined animals with a dorsal fin extending along the upper surface, continuing as a caudal fin around the tail and as a ventral fin to an atrium on the lower surface. They lack paired fins, and their body surface is covered by a smooth cuticle layer.

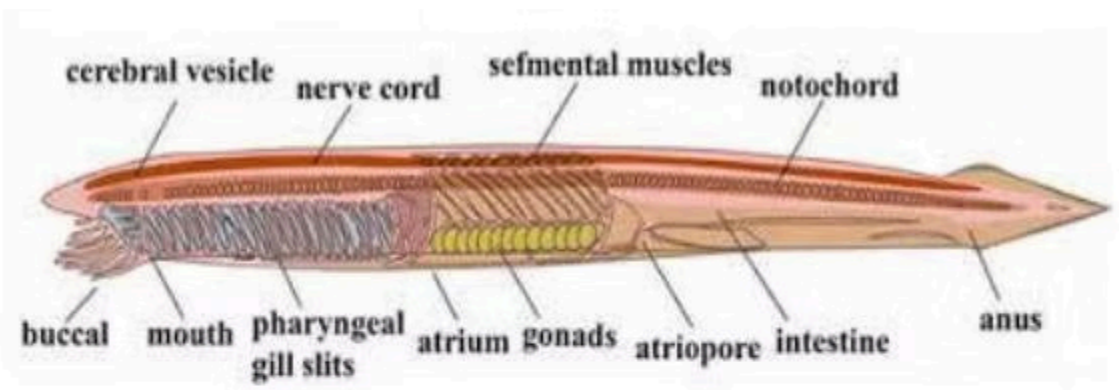
Habitat & Feeding

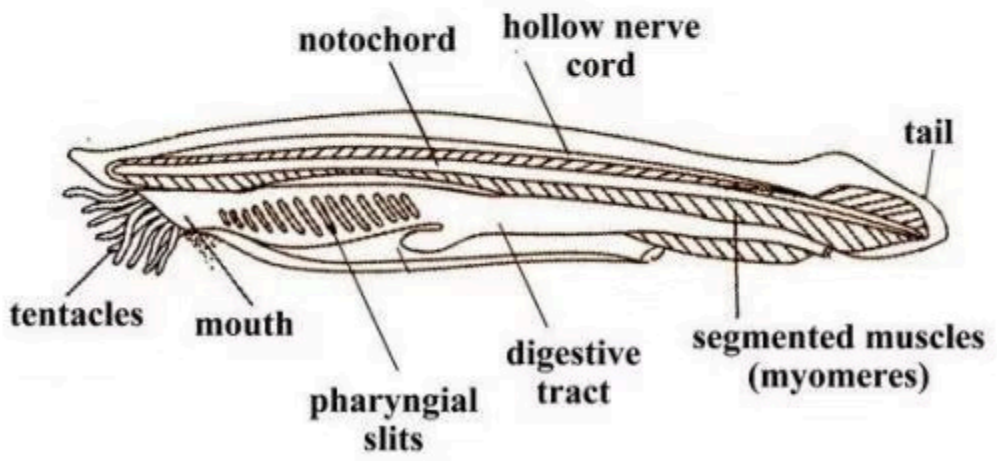
Lancelets are distributed worldwide along tropical and temperate coasts. They inhabit soft bottoms from sand to coarse shelly gravel in shallow coastal waters, often lying buried with only their mouths protruding to take in water laden with food.

Respiration

The gill serves for both feeding and gas exchange. After water passes through the gill slits, it reaches the atrium and exits through the atriopore. Excretory products and gametes also exit through this opening.

Cephalochordates and vertebrates share key characteristics including a hollow dorsal nerve cord, pharyngeal gill slits, and a notochord. However, in cephalochordates, the notochord is retained into adulthood rather than being replaced by vertebrae.





Study of Fishes

Fishes are members of the Chordate subphylum Vertebrata, possessing vertebrae that surround their spinal cord and a skull that protects the brain. They have traditionally been divided into superclasses based on whether they lack jaws and paired appendages (Agnathans) or possess these structures (Gnathostomates).

1

Agnathans (Jawless Fishes)

Class Myxini includes hagfishes, which are worm-like without scales. Their body is differentiated into head, trunk, and tail, reaching about 60cm in length. They have a mouth with four tentacles and 5-15 pairs of pharyngeal slits.

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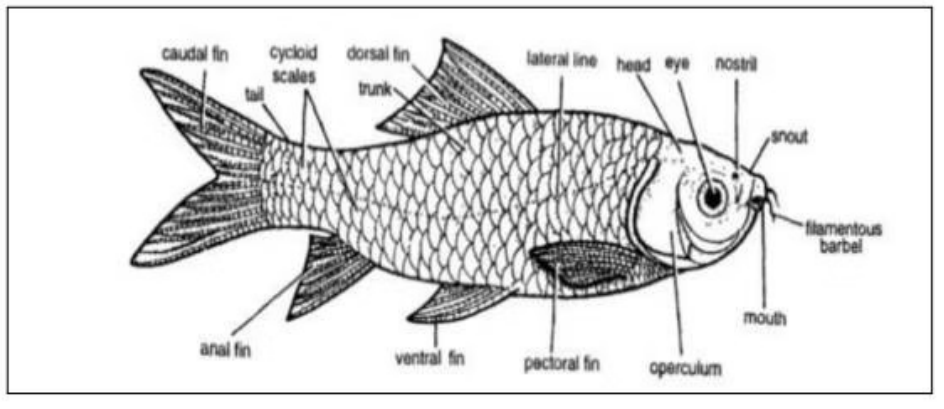
Osteichthyes (Bony Fishes)

Characterized by having bone in their skeleton, bony operculum covering gill openings, and swim bladder. Labeo rohita (rohu) is a common example with a compressed, fusiform body about 1 meter in length, covered with large overlapping cycloid scales.

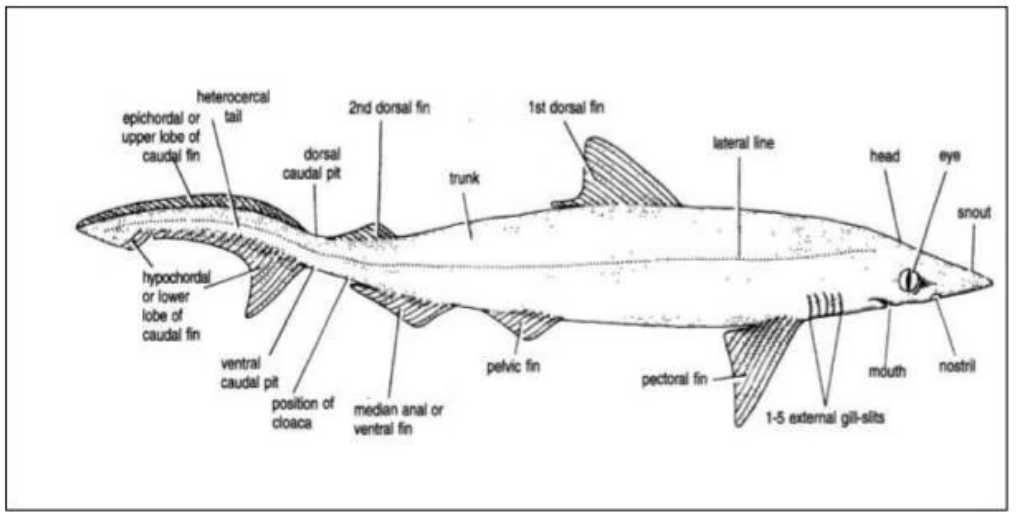
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Chondrichthyes (Cartilaginous Fishes)

Includes sharks, skates, and rays. They possess epidermal placoid scales and a cartilaginous endoskeleton. Scoliodon (dogfish) has a spindle-shaped body about 60cm long with a heterocercal tail and exhibits sexual dimorphism.



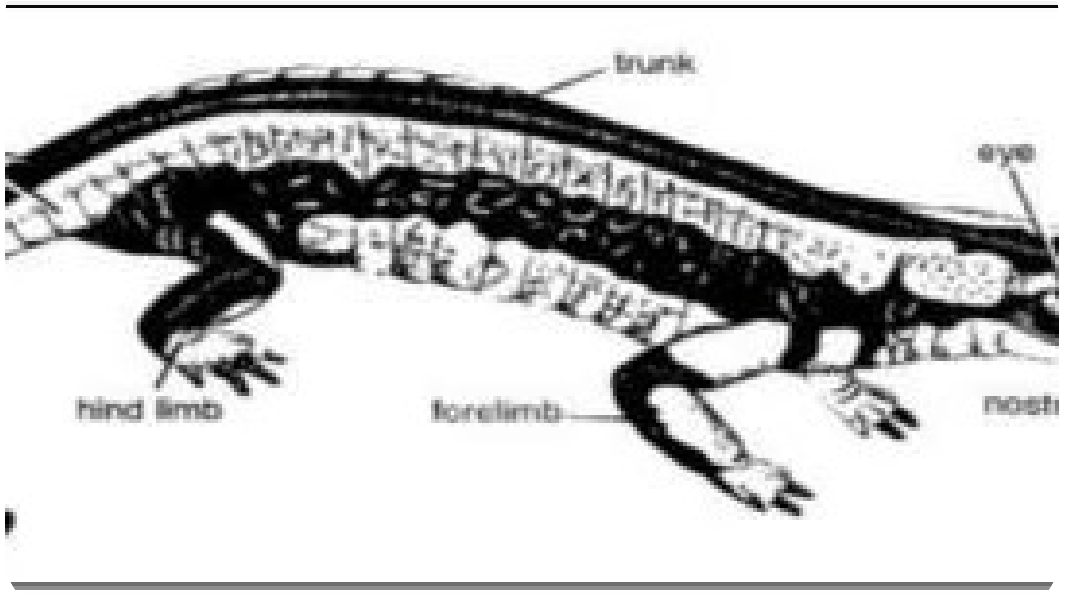
Labeo rohita (rohu) - A common carp with bluish back and silvery white underside, featuring a subterminal fringed-lipped mouth and homocercal tail.

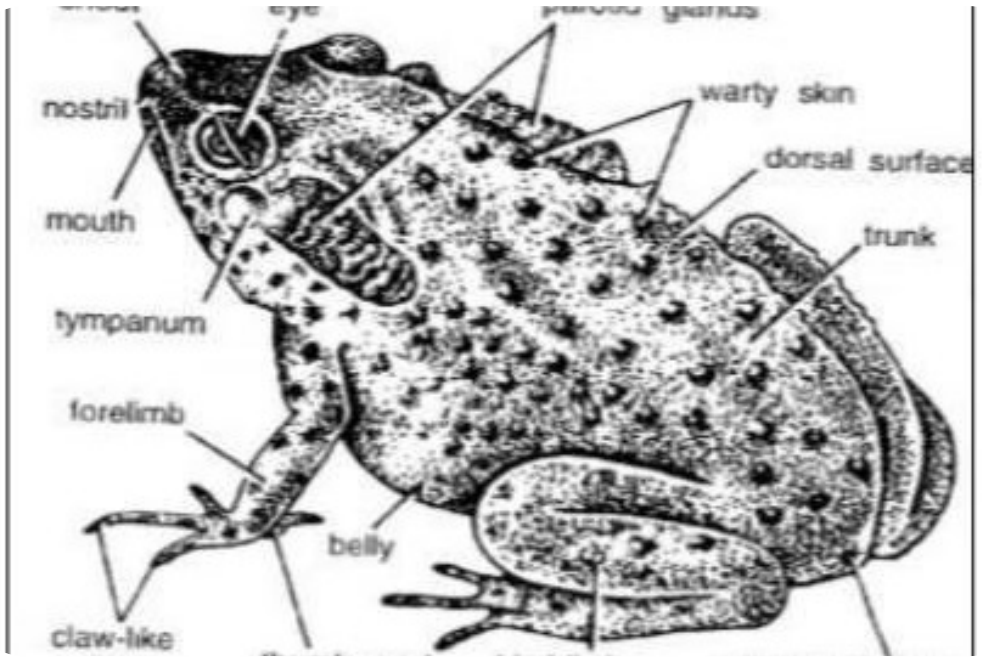


Scoliodon (Dogfish) - Features a dorso-ventrally compressed head with ventrally situated slit-like mouth and heterocercal tail. Males have claspers as intromittent organs.

Study of Amphibians

Amphibians occur on all continents except Antarctica but are absent from many oceanic islands. As tetrapods, they exhibit adaptations for life on land. Modern amphibians belong to three orders: Caudata (salamanders), Anura (frogs and toads), and Gymnophiona (caecilians).





Salamanders (Order Caudata)

Salamanders possess a tail throughout life and both pairs of limbs. They are lizard-like with females typically larger than males. Their body is divided into head, trunk, and tail with well-developed limbs. They have prominent eyes with movable eyelids and large parotid glands behind the head.

Toads (Order Anura)

Bufo (common toad) differs from frogs in having tough, dry, and warty skin with more poison glands than mucus glands. They are terrestrial and nocturnal, hiding under logs and stones during the day. Their body is divided into head and trunk, with short, poorly developed webbed limbs.

General characteristics of amphibians include skin with mucoid secretions lacking epidermal scales, feathers, or hairs; typically aquatic larvae that undergo metamorphosis to the adult form; and two atrial chambers in the heart. Their skin glands help keep them moist to prevent drying, while specialized cells called chromatophores are responsible for skin color and color change.

Study of Reptiles

Reptiles are air-breathing vertebrates with internal fertilization, amniotic development, and epidermal scales covering part or all of their body. The major groups include Turtles (Testudines), tuatara (Rhynchocephalia), lizards and snakes (Squamata), and crocodiles (Crocodylia).

Order Testudines

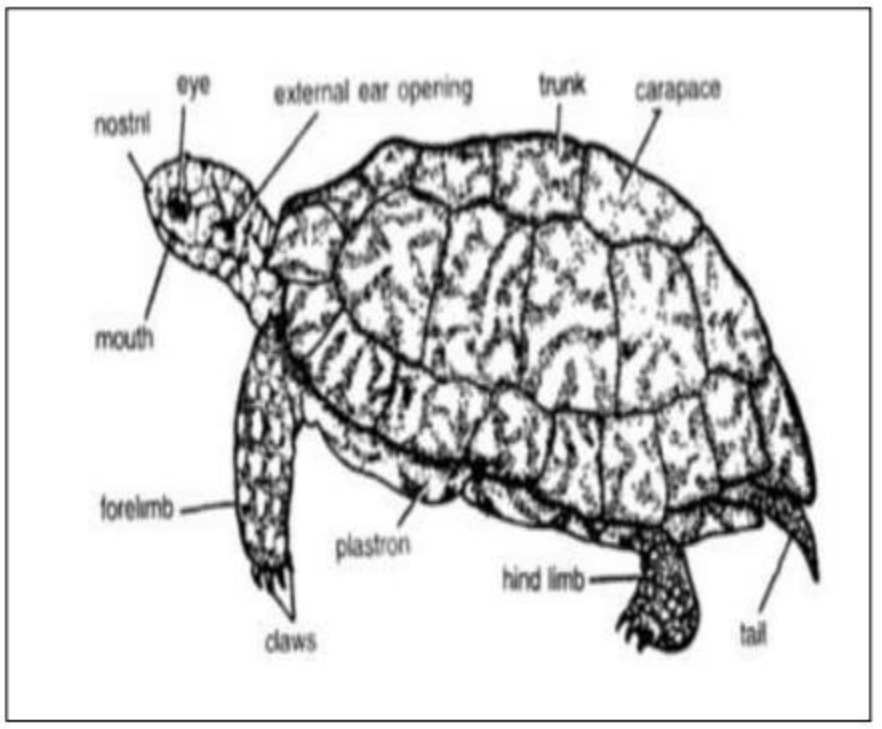
Giant turtles (Testudo) are found in fresh or salt water or on land. Their body is divided into head, trunk, and tail, with a leathery skin. The dorsal convex portion is called the carapace, and the flatter ventral portion is the plastron. They are uricotelic and oviparous, laying 5-100 eggs covered with soil.

Order Squamata

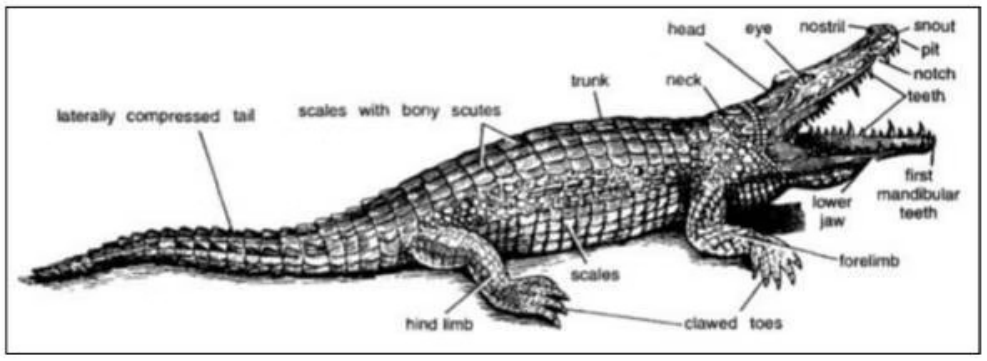
Hemidactylus (common house lizard) has worldwide distribution. They are nocturnal and hibernate during winter. Their body is divided into head, neck, trunk, and tail. The head is triangular with eyes lacking movable eyelids, and both forelimbs and hindlimbs are well-developed for walking.

Order Crocodilia

Crocodylus is found in Southern Asia, Africa, Australia, Central America, and India. They make 10-13 meter long tunnels below water level. Their snout is elongated for capturing food, with nostrils at the tip allowing breathing while mostly submerged. They are oviparous and display parental care.



Testudo (Giant Turtle)



Crocodylus (Mugger)

Study of Birds

Birds are unique among vertebrates because they have feathers on their bodies. They evolved from reptilian ancestors, as evidenced by scales on their legs and claws. Their major characteristics include adaptations for flight: wings, feathers, endothermy, high metabolic rate, and bones lightened by numerous air spaces.

Columba livia (Pigeon)

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The blue-rock pigeon has a grey plumage with metallic green and purple on breast and neck. Its head contains large eyes and slit-like nostrils, with a short, slender beak adapted for seed-eating. Forelimbs are modified into wings, while hindlimbs are adapted for bipedal locomotion.

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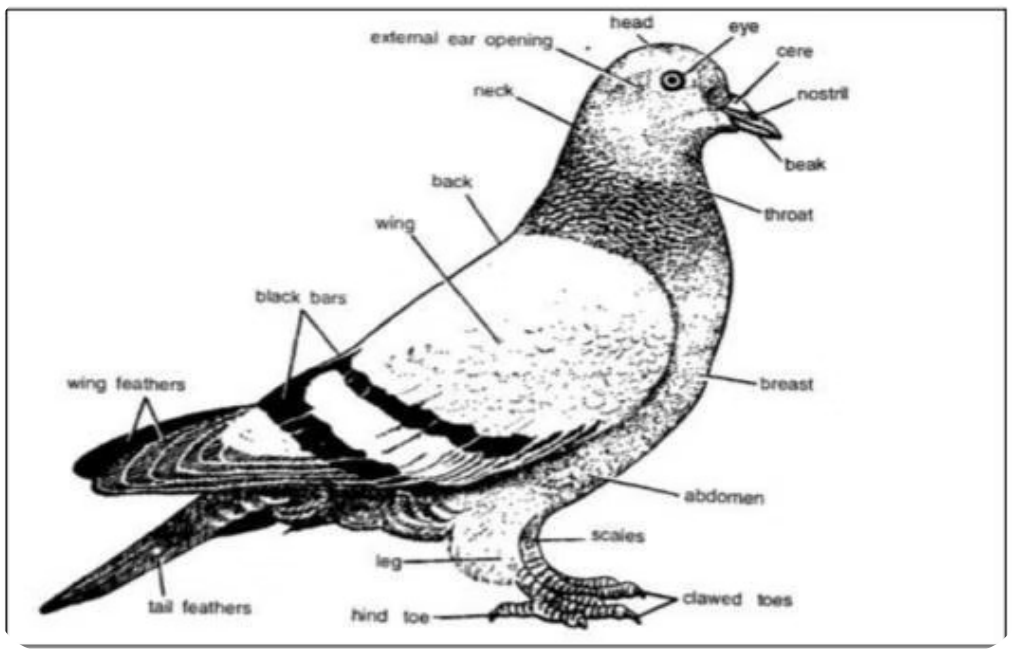
Psittacula euparia (Parrot)

The Indian parakeet has brilliant blue-green plumage with a massive, deeply-hooked red bill and a maroon patch on each shoulder. Its beak is stout, narrow, and hooked at the tip, adapted for fruit eating. Feet are zygodactylous for grasping, holding, and climbing.

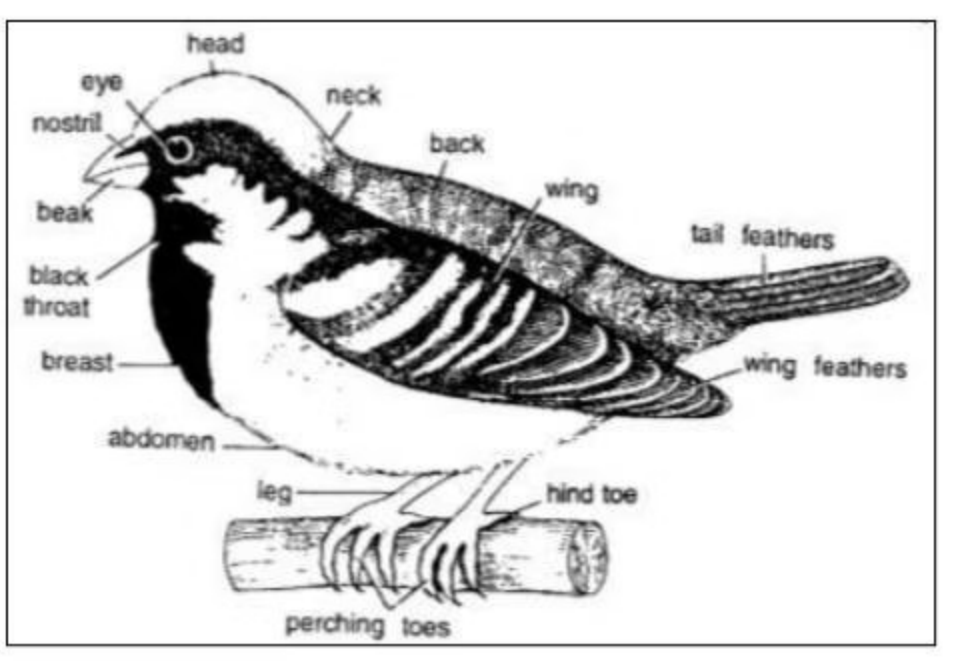
Passer domesticus (Sparrow)

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The common house sparrow exhibits sexual dimorphism: females are ash white, while males are earthy brown with blackish throat and breast and white abdomen. The beak is short and conical, adapted for seed eating. Feet are adapted for perching with 3 toes in front and 1 behind.



Columba livia (Pigeon)



Passer domesticus (Sparrow)

Birds have reduced skull bones with a keratinized bill replacing teeth. Their light bones support flight, and their forelimbs are modified into wings. Feet are covered with epidermal scutes formed by the fusion of several reptilian epidermal scales.

Study of Mammals

The beginning of the tertiary period about 70 million years ago marked the start of the "age of mammals." Mammalian characteristics evolved gradually over a period of 200 million years, including mammary glands, diaphragm, three middle-ear ossicles, and heterodont dentition.

Monotremes

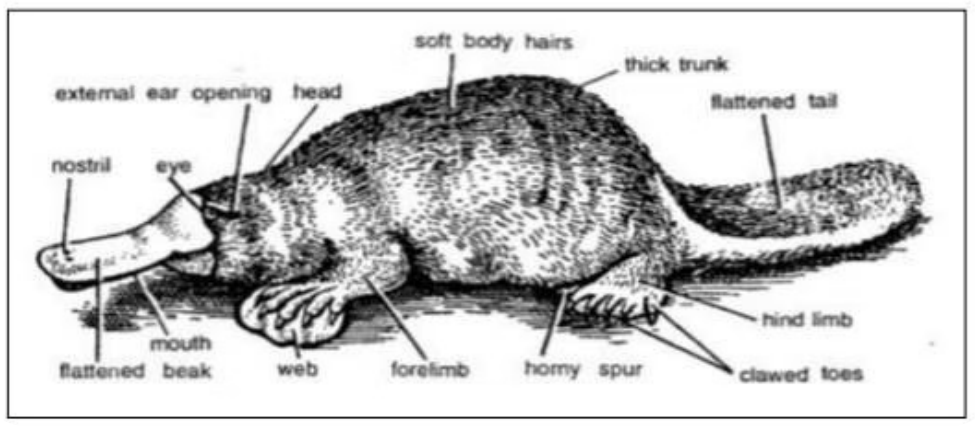
Egg-laying mammals like the duck-billed platypus (Ornithorhynchus). Native to Australia and Tasmania, they have soft hairs, no teeth in adults, and a flattened beak. Females make nests and lay 1-3 eggs, with young nursing by lapping milk secreted by mammary glands.

Metatheria (Marsupials)

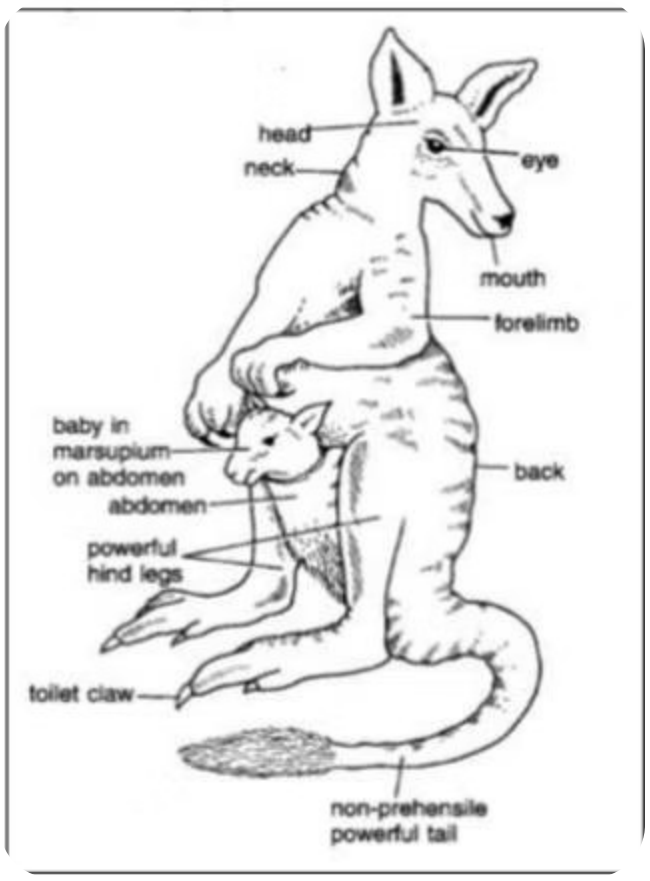
Pouched mammals like kangaroos (Macropus) where placenta doesn't form, so babies are born immature. Teats in the pouch provide milk for the young. Kangaroos have long hindlimbs for leaping and a powerful tail used for balance when resting.

Eutheria (Placental Mammals)

Warm-blooded animals with bodies covered in hair for insulation. They have two pairs of pentadactyl limbs adapted for various movements. The embryo develops inside the mother's body, with a placenta forming between fetus and uterine wall for nutrition.



Ornithorhynchus (Duck-bill)



Macropus (Kangaroo)

Mammals have a well-developed brain with large cerebral hemispheres and 12 pairs of cranial nerves. Their eyes are protected by movable eyelids, and their four-chambered heart ensures efficient circulation. The diversity of mammals reflects their successful adaptation to various ecological niches across the planet.